

Seokjoo Cho

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Champaign, IL

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EDUCATION

University of Illinois Urbana-Champaign

Ph.D. in Computer Science

Urbana, IL

Aug 2024 – Present

Chung-Ang University

B.E. in Computer Science and Engineering & B.S. in Mechanical Engineering

Seoul, South Korea

Mar 2016 – Aug 2022

PUBLICATIONS

*= equal contribution

- [1] Jiyu Hu, Seokjoo Cho*, Landon Johnson*, Kiran Hombal, Shreesha G. Bhat, Marcos K. Aguilera, Ramnatthan Alagappan, and Aishwarya Ganesan, “MEGALON: Efficient Data Sharing for Partly Coherent CXL Memory,” in *20th USENIX Symposium on Operating Systems Design and Implementation (OSDI '26)* (To Appear)
- [2] Kiet Tuan Pham*, Seokjoo Cho*, Sangjin Lee, Lan Anh Nguyen, Ipoom Jeong, Sungjin Lee, Nam Sung Kim, and Yongseok Son, “ScaleCache: A Scalable Page Cache for Multiple Solid-State Drives,” in *19th European Conference on Computer Systems (EuroSys '24)*
- [3] Seokjoo Cho, Sangjin Lee, Kiet Tuan Pham, Nguyen Lan Anh, Sunggon Kim, and Yongseok Son, “A Survey on Minimizing Lock Contention in Shared Resources in Linux Kernel,” in *International Conference on ICT Convergence (ICTC '22)*

RESEARCH EXPERIENCE

Research Assistant

University of Illinois Urbana-Champaign

Fall 2024 – Summer 2025

Urbana, IL

- **MEGALON.** Designed a software cache coherence mechanism that enables efficient data sharing on partly coherent CXL memory by leveraging shared-log-replicated metadata to scale to a large number of shared objects.

Undergraduate Researcher

Chung-Ang University

Dec 2021 – Jul 2024

Seoul, South Korea

- **ScaleCache.** Re-architected the Linux page cache to scale on systems with multiple high-performance NVMe SSDs, replacing centralized synchronization with a concurrent XArray structure and a cooperative writeback path that parallelizes dirty page flushing across application and flusher kthreads.

TEACHING EXPERIENCE

CS 341: System Programming

Teaching Assistant

Fall 2025 – Spring 2026

University of Illinois Urbana-Champaign

Linux System and its Application

Teaching Assistant

Fall 2022

Chung-Ang University

AWARDS

Top Excellence Award, 2022 CAU SW-AI Tech Fair (SW-AI Integrated Category)

School of Computer Science and Engineering, Chung-Ang University

Nov 2022

NRF President's Award, 4th X-corps Festival

National Research Foundation of Korea

Dec 2020

SERVICE

Shadow PC

- EuroSys 2026

Artifact Evaluation Committee Member

- FAST 2026, SOSP 2025

SKILLS

Programming Languages.

C, C++, Shell, Go, Rust, Java, Python, Linux kernel programming

Research Knowledge.

Ext4 file system, Btrfs, Linux page cache, NUMA, NVMe SSDs, RAID, CXL

Tools.

fio, perf, dmesg, gdb, stress-ng, [Data Access MONitor \(DAMON\)](#), QEMU